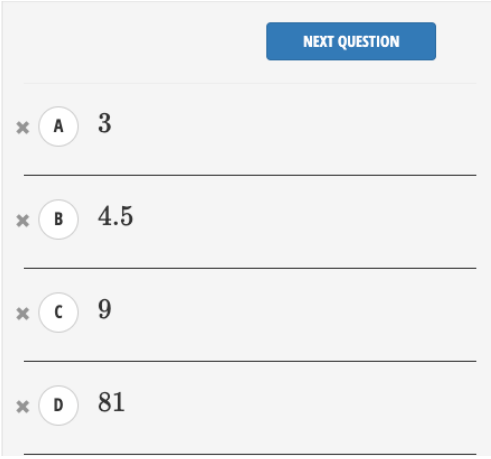
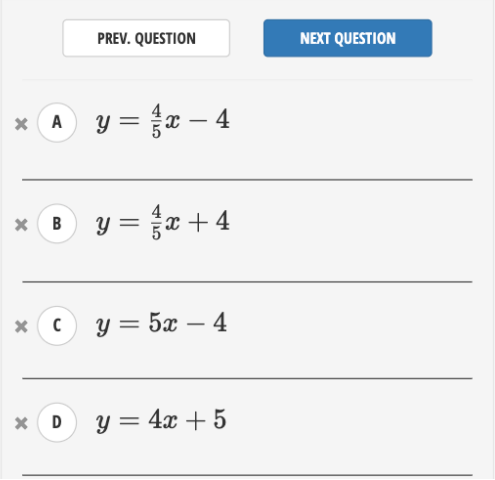
INNOVATOR TEST 8

MATH MODULE 1

1. The equation for a circle is $(x-1)^2+(y-2)^2=9$. What is the radius of the circle?



2. What is the equation of the line with a slope of $\frac{4}{5}$ and that passes through the point $(0,-4)$?



3.

$$y = -\frac{1}{5}x + 7$$
$$y = -\frac{1}{5}x - 3$$

How many solutions does the given system of equations have?

PREV. QUESTION | NEXT QUESTION

- A Zero
- B Exactly one
- C Exactly two
- D Infinitely many

4.

$$10x + 5 = 75$$
$$x + y^2 = 43$$

The solution to the given system of equations is (x, y). If $y > 0$, what is the value of y?

PREV. QUESTION | NEXT QUESTION

- A $\sqrt{35}$
- B 6
- C 7
- D 18

5. In a bag of 20 treats, 8 are chocolate, 9 are lollipops, and the rest are gumballs. Gumballs are p percent of the treats in the bag. What is the value of p?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

6.

Week	Bank Account
1	$200
2	$225
3	$250
4	$275
5	$300

The table shows how much money is in Aditya's bank account over time. Based on the table, how much money, in dollars, will Aditya have in his bank account on week 7?

PREV. QUESTION NEXT QUESTION

A 325

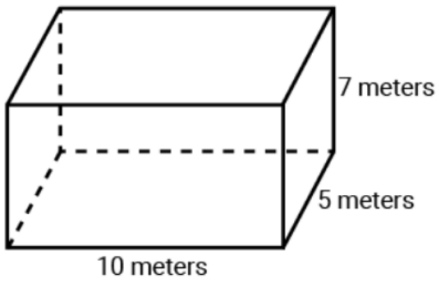
B 350

C 375

D 400

COME BACK LATER

7.



Note: figure not drawn to scale.

What is the volume of the rectangular prism shown, in cubic meters?

PREV. QUESTION NEXT QUESTION

A 88

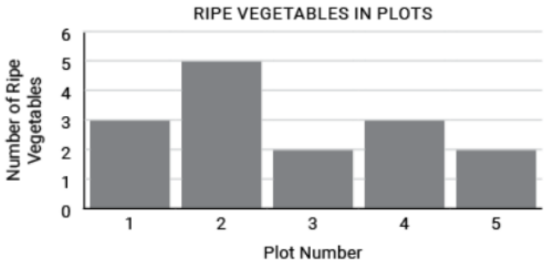
B 175

C 310

D 350

COME BACK LATER

8.



A gardener wants to know how many vegetables are ripe on his farm. Out of the 100 different square plots in his garden, he randomly selects five of them and counts how many ripe vegetables there are in each plot. The results are shown in the bar graph. Based on the mean of the sample, which of the following is the best estimate for the number of ripe vegetables in the entire farm?

PREV. QUESTION NEXT QUESTION

A) 3

B) 30

C) 300

D) 500

COME BACK LATER

9. A quadrilateral has three angles measuring 62°, 136°, and 31°. What is the measure of the fourth angle, in degrees?

PREV. QUESTION NEXT QUESTION

COME BACK LATER

10. Given the equations $3x + 4y = 20$ and $y = x(-x - 4)$, how many solutions (x, y) exist for the system?

PREV. QUESTION NEXT QUESTION

A) 0

B) 1

C) 2

D) 4

11. In triangle DEF, $\sin(E) = \frac{60}{75}$ and angle D is a right angle. What is the value of $\sin(F)$?

PREV. QUESTION | NEXT QUESTION

- ✖ A $\frac{60}{75}$
- ✖ B $\frac{45}{75}$
- ✖ C $\frac{45}{60}$
- ✖ D $\frac{60}{45}$

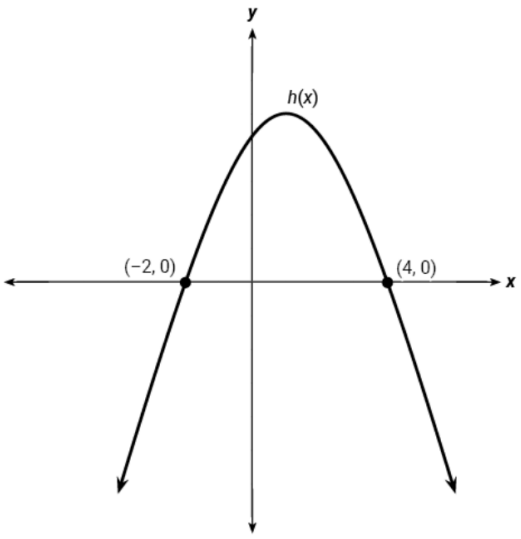
12. $\frac{1}{3}w + \frac{2}{3}w = \frac{1}{4}(w - \frac{1}{2}) + 1w$

What is the solution to the given equation?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

13.



The graph of $y = h(x)$ is shown in the xy-plane. Which of the following could be the equation of $h(x)$?

PREV. QUESTION | NEXT QUESTION

A $h(x) = -(x + 2)(x - 4)$

B $h(x) = -(x - 2)(x + 4)$

C $h(x) = (x + 2)(x - 4)$

D $h(x) = (x - 2)(x + 4)$

COME BACK LATER

14.

$$x^2 - 6x + 7 = 0$$

Which values satisfy the given equation?

PREV. QUESTION | NEXT QUESTION

A $x = -3 + \sqrt{2}$ and $x = -3 - \sqrt{2}$

B $x = -1$ and $x = -7$

C $x = 1$ and $x = 7$

D $x = 3 + \sqrt{2}$ and $x = 3 - \sqrt{2}$

15. Water is being drained from a swimming pool. The amount of water $w(h)$, in thousands of liters, after h hours can be found using the function $w(h) = -6.5h + 39$. How long will it take, in hours, before the swimming pool is empty?

PREV. QUESTION | NEXT QUESTION

× A 5

× B 6

× C 7

× D 32.5

16. $$c(t) = 4{,}000(0.5)^{\frac{t}{5{,}730}}$$

The function $c(t)$ is used to determine the amount of carbon–14 left in a sample t years after the sample is initially measured. How many years will it take the sample to contain half as much carbon–14 from when it was initially measured?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

17. The function $g(x)$ is defined as $g(x) = ax^2 + 3x$, and $g(4) = -20$. What is the value of $g(3)$?

PREV. QUESTION | NEXT QUESTION

× A -9

× B -2

× C $\frac{4}{5}$

× D 18

18. $3x^3 - 9x^2 - 54x$

One of the factors of the given expression is $x + k$, and k is a constant. What is the smallest possible value of k?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

19. $\frac{(a^3)^{-2}b^5c^{\frac{3}{2}}c^{-1}}{a^6b^{-5}c^{\frac{1}{2}}}$

Which of the following is equivalent to the expression shown above?

PREV. QUESTION | NEXT QUESTION

A 1

B $\frac{a^{12}}{b^{10}}$

C $\frac{b^{10}}{a^{12}}$

D $\frac{b^{10}}{a^{12}c^2}$

20. $2x^2 - y = 293$

$3x = y - 6$

When graphed on the xy-plane, the graphs of the equations in the system intersect at the point (x, y). What is a possible value of x?

PREV. QUESTION | NEXT QUESTION

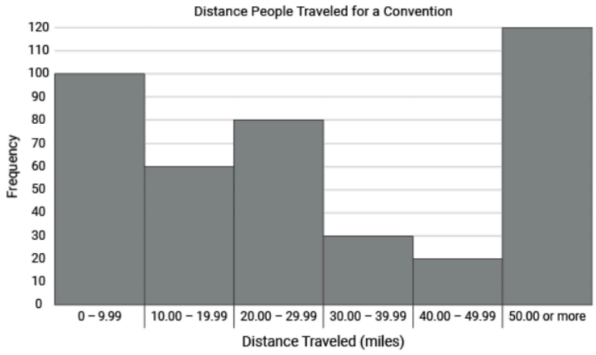
A -28.5

B -12

C 6

D 13

21.



The histogram summarizes the total distance people traveled to visit a city for a convention. If the average distance traveled for anyone who traveled 50 miles or more is 57.82 miles, approximately what is the greatest possible mean distance traveled by all who attended the convention?

PREV. QUESTION | NEXT QUESTION

× A 24.14

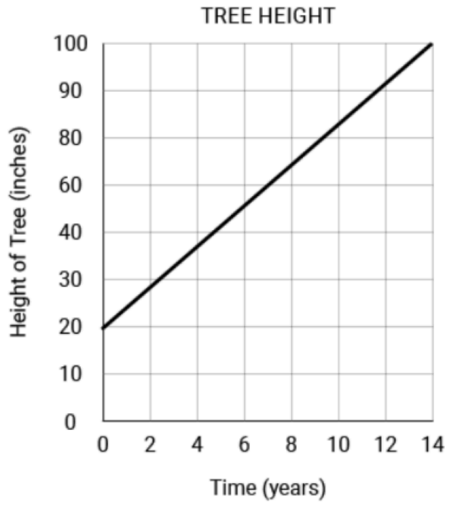
× B 29.97

× C 33.50

× D 40.63

COME BACK LATER

22.



The graph shows the height of a tree after a certain number of years. Let $h(t)$ represent the height of the tree after t years. Assuming the function is always linear, for which value of t will $h(t)$ equal 260?

PREV. QUESTION | NEXT QUESTION

× A 42

× B 65.5

× C $1,371.4$

× D $1,505.7$

COME BACK LATER

MATH MODULE 2 (HARD)

1. Angela buys 2 cookies and 3 brownies for a total of $\$19$. Vicky buys 4 cookies and 1 brownie for a total of $\$13$. If c represents the cost of a cookie and b represents the cost of a brownie, which system of equations can determine the cost of one cookie and one brownie?

NEXT QUESTION

- A $c + b = 5$, $6c + 4b = 32$
- B $2c + 3b = 19$, $4c + b = 13$
- C $5c = 19b$, $5b = 13c$
- D $6c = 19$, $4b = 13$

2. $$y = \frac{1}{3}x$$
$$y = -2x - 7$$

The given system of equations is graphed in the xy-plane, and the two lines intersect at the coordinate point $(-3, y)$. What is the value of y?

PREV. QUESTION NEXT QUESTION

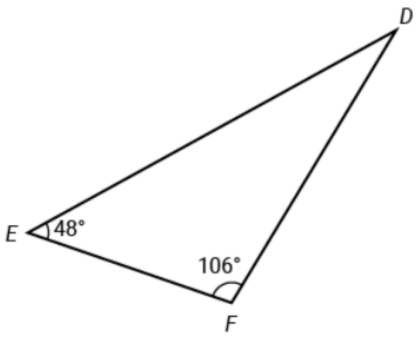
- A -9
- B -1
- C 1
- D 13

3. Which expression is equivalent to $x^2 + 5x - 6$?

PREV. QUESTION | NEXT QUESTION

- A $(x + 2)(x - 3)$
- B $(x - 2)(x + 3)$
- C $(x + 1)(x - 6)$
- D $(x - 1)(x + 6)$

4.



Note: Figure not drawn to scale

Given triangle DEF, what is the measure of angle D, in degrees?

PREV. QUESTION | NEXT QUESTION

- A 26
- B 58
- C 154
- D 206

COME BACK LATER

5. A circle in the xy-plane has its center at $(-3, 7)$, and the point $(1, 7)$ lies on the circle. Which equation represents this circle?

PREV. QUESTION | NEXT QUESTION

- A $(x - 3)^2 + (y + 7)^2 = 4$
- B $(x + 3)^2 + (y - 7)^2 = 4$
- C $(x - 3)^2 + (y + 7)^2 = 16$
- D $(x + 3)^2 + (y - 7)^2 = 16$

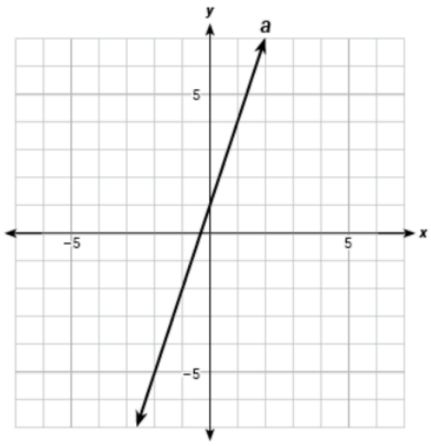
6. $\frac{x+7}{2(x+1)} = \frac{3}{5}$

What is the solution to the given equation?

PREV. QUESTION | NEXT QUESTION

- A -33
- B -3
- C 5
- D 29

7.

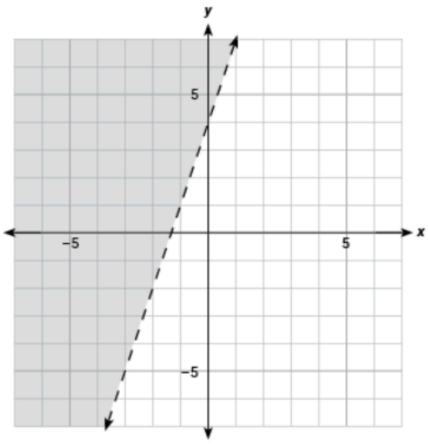


The graph of line a is shown, and the equation of line a is $y = 3x + 1$. Line b is parallel to line a and crosses through the points $(0, 5)$ and $(k, -7)$. What is the value of k?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

8.



A linear inequality is graphed on the xy–coordinate plane.

Which coordinate point is a solution to the graphed inequality?

PREV. QUESTION | NEXT QUESTION

× A $(-3, -4)$

× B $(-2, -3)$

× C $(0, 4)$

× D $(1, 6)$

COME BACK LATER

9.

$$2x^2 - 40x + 182 = 0$$

What is one of the solutions to the given equation?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

10.

Data set X : $25, 32, 11, 42, 33, 16, 27, 39, 54$

Data set X is shown. If 54 is removed from the data set, which of the following is true?

PREV. QUESTION | NEXT QUESTION

× A The mean and the median do not change.

× B The mean and the median change by the same amount.

× C The change to the median is greater than the change to the mean.

× D The change to the mean is greater than the change to the median.

11. $2(37x + 1) = 37(2x + 1)$

How many solutions does the given equation have?

PREV. QUESTION | NEXT QUESTION

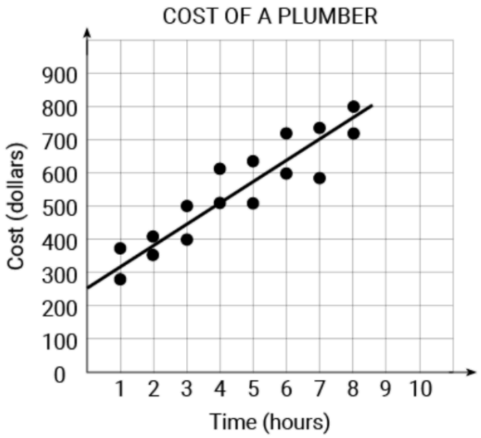
- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

12. The function $d(t)$ is defined as $d(t) = 6t + 14$. What is the value of $d(1.5) + d(2.5)$?

PREV. QUESTION | NEXT QUESTION

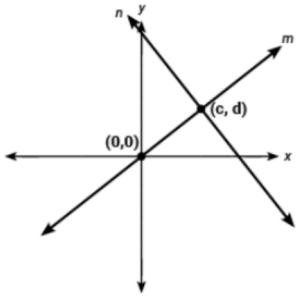
- A. 23
- B. 29
- C. 38
- D. 52

13.



The scatterplot shows the comparison for the time and the cost to hire a plumber. At what hour is the predicted cost the closest to an actual cost to hire a plumber?

14.



Two linear equations m and n are graphed on the xy–coordinate plane, as shown. If line m is perpendicular to line n, what is the value of the y–intercept of line n?

PREV. QUESTION NEXT QUESTION

× A $d - \frac{c^2}{d}$

× B $d + \frac{c^2}{d}$

× C $d - \frac{d^2}{c}$

× D $d + \frac{d^2}{c}$

15. In certain conditions, filamentous fungi will grow exponentially, tripling in size every day. If, on day 1, the fungi are each 4 mm in length, which of the functions below represents the length on the nth day?

PREV. QUESTION NEXT QUESTION

× A $y = 3(4)^n$

× B $y = 3(4)^{n-1}$

× C $y = 4(3)^n$

× D $y = 4(3)^{n-1}$

16. $2x^2 - 27x + d = 0$

In the given equation, d is a positive integer constant. The equation has no real solutions. What is the smallest possible value for d

PREV. QUESTION | NEXT QUESTION

× A 91

× B 92

× C 93

× D 94

17. $m^6 = (n + 5p)^3$

The given equation relates the variables m, n, and p for all real numbers > 0. Which equation correctly expresses n in terms of m and p?

PREV. QUESTION | NEXT QUESTION

× A $n = \sqrt[3]{m^3 - 5p}$

× B $n = m^2 - \sqrt[3]{5p}$

× C $n = m^2 - 5p$

× D $n = m^3 - 5p$

18.

Number of shirts	Cost of Shirts	Delivery Cost	Tax	Total Cost
6	$210.00	$25.20	$23.52	$258.72
15	$525.00	$63.00	$58.80	$646.80
21	$735.00	$88.20	$82.32	$905.52

The table above shows the price for ordering different quantities of shirts. A delivery fee is calculated as a percentage of the base cost of the shirts. Then, a tax is applied to the combined cost of the shirts and the delivery. If p is the percentage added for delivery and q is the percentage added for taxes, what is the value of $p - q$?

PREV. QUESTION NEXT QUESTION

× A 0.8

× B 2

× C 6.67

× D 23.2

COME BACK LATER

19. In the right triangle ABC, angle C is a right angle. If $\cos(A) = \frac{x-k}{x+k}$, where k is a constant and x is the side length of BC, what is the value of x, in terms of k?

PREV. QUESTION NEXT QUESTION

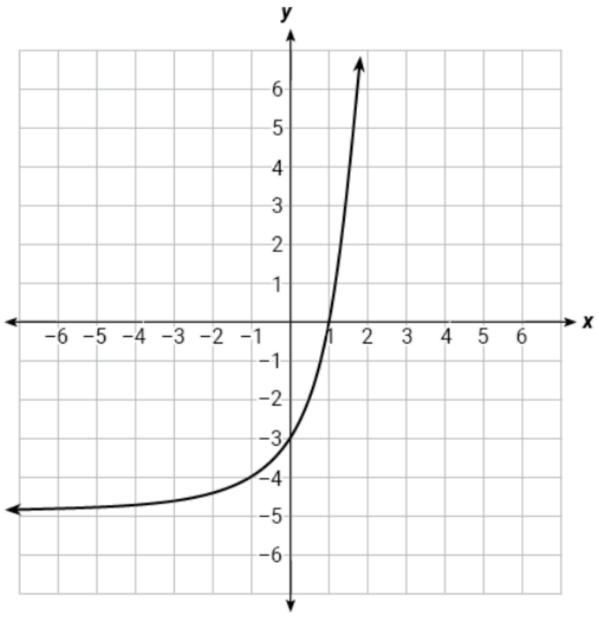
× A $2k$

× B $4k$

× C $2k^2$

× D $4k^2$

20.



The exponential function $f(x)$ is graphed on the xy–plane. The function $g(x)$ (not shown) is defined such that $g(x) = f(x-1) + k$. If $g(0) = 1$, what is the value of k?

PREV. QUESTION | NEXT QUESTION

- A -5
- B -4
- C 4
- D 5

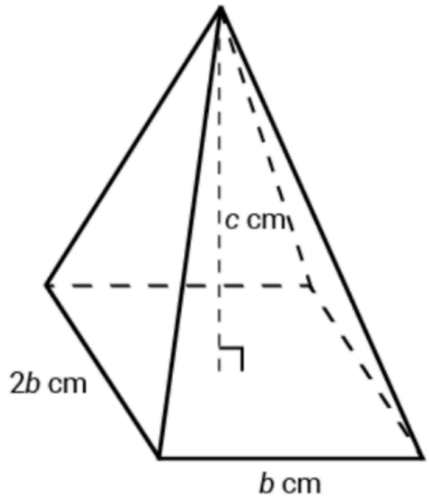
COME BACK LATER

21. A cruise ship charges $\$4$ per soda for the first 19 sodas and then charges $\$2.50$ for each additional soda. If $c(s)$ is the total cost, in dollars, for s sodas and $s > 19$, what is the equation of $c(s)$?

PREV. QUESTION | NEXT QUESTION

- A $c(s) = 2.5s + 19$
- B $c(s) = 2.5s + 28.5$
- C $c(s) = 2.5s + 76$
- D $c(s) = 6.5s + 47.5$

22.



Note: Figure not drawn to scale.

A rectangular pyramid has a base length measuring b centimeters, a base width measuring $2b$ centimeters, and a height measuring c centimeters, as shown. A similar rectangular pyramid is $\frac{2}{3}$ the size of the given rectangular pyramid. What is the volume of the smaller rectangular pyramid?

PREV. QUESTION

A $\frac{2b^2c}{3}$

B $\frac{4b^2c}{9}$

C $\frac{8b^2c}{27}$

D $\frac{16b^2c}{81}$

COME BACK LATER